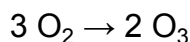
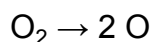


Ozone (O_3) in the atmosphere protects against harmful UV radiation. The overall reaction for ozone formation is



and proceeds in two steps. Step 1 is initiated when molecular oxygen (O_2) absorbs UV light and splits into two oxygen atoms:



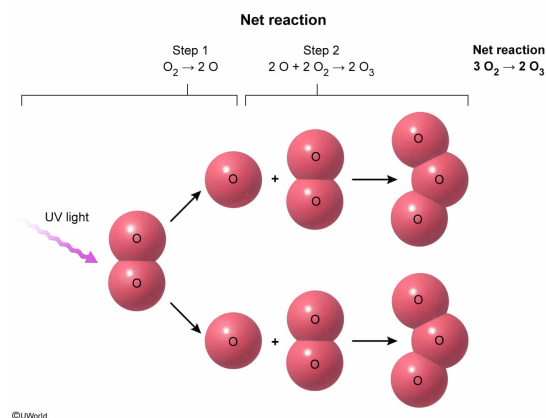
Which of the following second steps would yield the net reaction for ozone formation?

- ☐ A. $\text{O} + \text{O}_2 \rightarrow \text{O}_3$ [11%]
- ✓ ☒ B. $2 \text{O} + 2 \text{O}_2 \rightarrow 2 \text{O}_3$ [74%]
- ☐ C. $\text{O}_2 + 2 \text{O} \rightarrow 2 \text{O}_3$ [11%]
- ☐ D. $\text{O} + 2 \text{O}_2 \rightarrow \text{O}_3$ [3%]

Correct

74% answered correctly

Explanation:



The **law of conservation of mass** says that in a chemical reaction, atoms **are neither created nor destroyed**: they can only be rearranged. Therefore, in any chemical reaction the **same number** of each type of atom must be present on the left and right sides of the reaction arrow. Reactions may proceed in multiple steps, producing intermediates along the way. If the reaction **goes to completion** (the maximum possible amount of product is formed), all **intermediate species** must be **consumed** in subsequent steps to form the final products.

In the formation of ozone (O_3), first, one molecule of oxygen (O_2) is split into two oxygen atoms as intermediates ($\text{O}_2 \rightarrow 2 \text{O}$). The oxygen atoms are then consumed as each reacts with an oxygen molecule to form ozone. For the reaction to go to completion, the two oxygen atoms must combine with two O_2 molecules to form two ozone molecules ($2 \text{O} + 2 \text{O}_2 \rightarrow 2 \text{O}_3$) in the **second step**. This yields the net reaction $3 \text{O}_2 \rightarrow 2 \text{O}_3$.

(Choice A) $\text{O} + \text{O}_2 \rightarrow \text{O}_3$ correctly shows the reaction of one oxygen atom with one O_2 molecule to form one ozone molecule. However, if this were the second step it would not yield the correct net reaction. Instead, the resulting net reaction would be $2 \text{O}_2 \rightarrow \text{O}_3 + \text{O}$.

(Choice C) $\text{O}_2 + 2 \text{O} \rightarrow 2 \text{O}_3$ has four oxygen atoms on the left and 6 oxygen atoms on the right, so it is unbalanced. If it were the second step, the resulting net reaction would be $2 \text{O}_2 \rightarrow 2 \text{O}_3$, which is also unbalanced.

(Choice D) $\text{O} + 2 \text{O}_2 \rightarrow \text{O}_3$ has five oxygen atoms on the left side and only three on the right, leaving two oxygen atoms unaccounted for. If it were the second step, the resulting net reaction would be $3 \text{O}_2 \rightarrow \text{O}_3 + \text{O}$, which is unbalanced.

Educational objective:

According to the law of conservation of mass, atoms are neither created nor destroyed in a chemical reaction. The same number of each type of atom must be present on both the left and right sides of a reaction arrow. Multistep reactions that go to completion must consume all intermediates produced before or during the final step.

Time spent:0 sec

Subject:
General Chemistry

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Interactions of Chemical Substances